

Control as Inference: Path Integrals for Stochastic Optimal Control

gpt-5-codex

A controlled diffusion is a distribution over trajectories, not a single path. This note asks six questions about the path-weight view of stochastic optimal control: what a deterministic planner misses, what exponential tilt interacts with a change of path measure, what must be normalized, what change of variable linearizes the Hamilton–Jacobi–Bellman equation, how the resulting path integral can be estimated without enumerating paths, and what fails when noise and control are mismatched or the noise vanishes. The running example is a scalar continuous-diffusion plant with additive noise, matched control and noise directions, positive noise strength, and quadratic control cost.

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1. What does a single minimizing trajectory miss?

Question: For a plant whose state is driven by noise, a deterministic planner compares single trajectories. What object must a stochastic planner compare instead, and why can two controls that produce the same nominal path have different costs?

Running example. A scalar state x_t follows the controlled diffusion

$$dx_t = (f(x_t) + Gu_t) dt + \sqrt{\varepsilon} dW_t, \quad \varepsilon > 0, \quad (1)$$

where u_t is the control, G maps the control into the state direction, and W_t is a one-dimensional Wiener process. The finite-horizon cost is

$$J(u) = \mathbb{E}_u \left[\int_0^T \left(\ell(x_t) + \frac{1}{2} u_t^T R u_t \right) dt + \Phi(x_T) \right], \quad (2)$$

with ℓ the running state cost, Φ the terminal cost, and R positive definite. The discrete-time bridge used in the figures is

$$x_{k+1} = x_k + \Delta t(f_k + Gu_k) + \sqrt{\varepsilon \Delta t} \xi_k, \quad \xi_k \sim \mathcal{N}(0, 1). \quad (3)$$

Facts to reason with.

- The expectation in Equation 2 is over paths of Equation 1, not over a single trajectory.
- The control may be a feedback law, so u_t can depend on the current state.
- The noise is additive: changing u_t changes the drift, not the diffusion coefficient.
- A nominal path is obtained by setting the noise term in Equation 3 to zero.

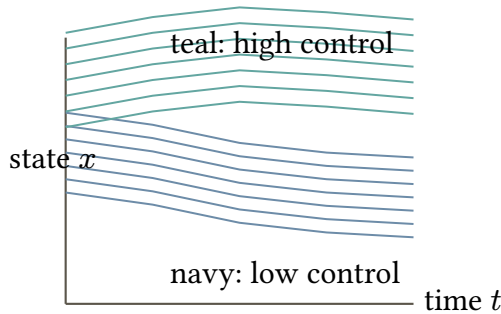


Figure 1: Two bundles of sample paths from Equation 3. Inspect how each bundle spreads around its nominal path and how the two bundles differ.

2. What exponential tilt interacts with the change of path measure?

Question: The controlled path measure differs from the uncontrolled one by a Radon–Nikodym derivative. What exponential tilt of the path cost Equation 2 makes the control-dependent part of that derivative and the running control cost interact, and what relation between R and G is required for the interaction to depend only on the noise?

Path measures. Let $\mathbb{P}_0$ be the path measure of the uncontrolled dynamics

$$dx_t = f(x_t) dt + \sqrt{\varepsilon} dW_t, \quad (4)$$

and let $\mathbb{P}_u$ be the path measure of the controlled dynamics Equation 1. For the additive noise in Equation 1, the Radon–Nikodym derivative is

$$\frac{d\mathbb{P}_u}{d\mathbb{P}_0}(\omega) = \exp\left(\int_0^T \frac{(Gu_t)^T}{\sqrt{\varepsilon}} dW_t - \frac{1}{2} \int_0^T \frac{\|Gu_t\|^2}{\varepsilon} dt\right). \quad (5)$$

Facts to reason with.

- The running control cost in Equation 2 is $\frac{1}{2}u_t^T Ru_t$.
- The Girsanov factor Equation 5 contains Gu_t and the noise strength ε .
- In the scalar running example, $G = 1$, $R = 1$, and $\varepsilon > 0$.
- The terminal cost $\Phi(x_T)$ depends on the path and does not contain u_T explicitly.

3. What must be normalized?

Question: A positive path weight assigns a number to each trajectory. What operation turns that weight into a distribution over trajectories, and what does the resulting normalization constant represent when the weight is built from a control cost?

Path weights. Let $w(\omega) > 0$ be a candidate weight on paths. A constant Z makes $p(\omega) = \frac{w(\omega)}{Z}$ a probability density over paths. The path space here is the space of continuous trajectories on $[0, T]$, so sums over finitely many paths are replaced by an integral over an infinite-dimensional space.

Facts to reason with.

- Z is a single positive number, not a function of the path.
- The normalized weight p assigns more mass to paths with larger w .
- The running example keeps the same state cost, terminal cost, control cost, and noise strength as Equation 2 and Equation 5.
- A control law changes the distribution of paths, so two different controls generally produce two different normalized weights.

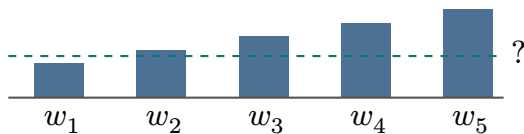


Figure 2: Positive path weights w_i for five trajectories. Inspect the bars and the dashed level marked ?; the question asks what the missing constant means, not which bar is largest.

4. What change of variable linearizes the HJB equation?

Question: The Hamilton–Jacobi–Bellman equation Equation 6 is nonlinear because of the quadratic term in the gradient of the value function. What change of variable removes that nonlinearity, and how is the optimal control recovered from the transformed variable?

Value function. Let $v(t, x)$ be the value function for Equation 2 and Equation 1 on $[t, T]$. The Hamilton–Jacobi–Bellman equation is

$$-\partial_t v(t, x) = \min_u \left\{ \ell(x) + \frac{1}{2} u^T R u + (f(x) + G u)^T \nabla v(t, x) + \frac{\varepsilon}{2} \Delta v(t, x) \right\}, \quad (6)$$

with terminal condition

$$v(T, x) = \Phi(x). \quad (7)$$

Facts to reason with.

- ∇v and Δv are taken with respect to the state x .
- The minimization over u is inside the equation; the minimizer is a function of ∇v .
- In the scalar running example, $G = 1$, $R = 1$, and $\varepsilon > 0$.
- The terminal condition Equation 7 is fixed by Φ .

5. How can the path integral be estimated without enumerating paths?

Question: The path integral involves an expectation over an infinite-dimensional path space. If a simulator can draw trajectories from a reference process, which reference drift makes the sample weights computable for the running example, and which part of the weight determines how quickly the estimator's variance grows with the horizon?

Reference process. Suppose the simulator can draw paths from

$$dx_t = b(x_t) dt + \sqrt{\varepsilon} dW_t \quad (8)$$

for a chosen drift b . An expectation under one path measure can be rewritten as an expectation under another by multiplying each sampled path by a Radon–Nikodym weight.

Facts to reason with.

- The reference drift b in Equation 8 is a modeling choice.
- The noise strength ε and the terminal time T are fixed in the running example.
- A Monte Carlo estimate from N independent paths has error that decreases like $\frac{1}{\sqrt{N}}$.
- The variance of the estimate depends on the distribution of the path weights, not only on N .

6. What breaks when matching fails or noise vanishes?

Question: The matched case has $R = G^T G$ and $\varepsilon > 0$. Remove one condition at a time: when the control directions and noise directions no longer match, what fails? When $\varepsilon \rightarrow 0$, what fails? Which failure belongs to the path-weight construction, and which belongs to the Monte Carlo estimator?

Two perturbations. Keep the plant Equation 1, the cost Equation 2, and the Girsanov factor Equation 5. The matched case is $R = G^T G$ with $\varepsilon > 0$. The mismatched case keeps $\varepsilon > 0$ but allows $R \neq G^T G$. The small-noise case keeps $R = G^T G$ but takes $\varepsilon \rightarrow 0$.

Facts to reason with.

- The control cost uses R ; the path-measure change uses G .
- The Girsanov factor Equation 5 contains $\frac{1}{\sqrt{\varepsilon}}$ and $\frac{1}{\varepsilon}$.
- As $\varepsilon \rightarrow 0$, the diffusion term in Equation 1 vanishes and the plant approaches a deterministic system.
- The normalized path weight from section 3 is still defined for every $\varepsilon > 0$.

7. Sources and further reading

No external sources have been consulted yet; this note makes no external claims.